Running LatentMAS in one engine

All of LatentMAS inside a modified vLLM · Qwen3-14B · one H200

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29 October 2026

How LatentMAS runs today

LatentMAS is four copies of the same model working on one question. Three helpers "think silently" for a few steps; the fourth, the Judger, reads those thoughts and writes the answer.

The authors run this in two separate programs:

- A research library (Hugging Face) does the silent thinking. It is easy to modify but slow, and handles one batch at a time.
- A fast serving program (vLLM) writes the Judger's answer.
- So the model is loaded into memory **twice**, and the thoughts are handed from one program to the other and re-read.

vLLM does not support silent thinking out of the box. The authors' README says they modified parts of vLLM for this; that change is not in their public code.

What we built

A modified copy of vLLM that does **all** of LatentMAS by itself.

- New mode: instead of picking the next word, the model feeds its own internal state back in as the next input. That is the silent thinking.
- Each helper's instructions are added to the same running request, so the Judger picks up exactly where the helpers left off.
- One copy of the model in memory, no hand-off, no re-reading.
- Many questions can be processed side by side, including the silent thinking.
- If memory runs out and a question gets paused, its silent thoughts are saved so it can resume correctly. We tested this on purpose.

Grade-school math, one question at a time

Qwen3-14B, 50 GSM8K questions (the two-program version ran the first 30). Time is the typical (middle) value.

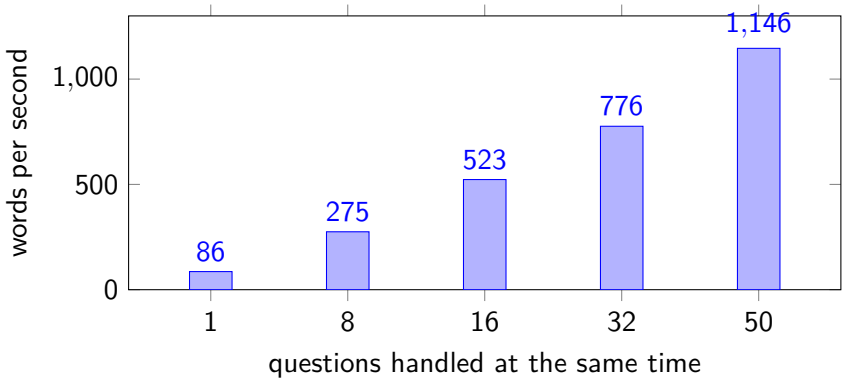
how it was run	correct	time per question	words per second
Paper's way (research library only)	49/50	13.3 s	24
Authors' two-program version	30/30	5.1 s	56
Ours (one engine)	**49/50**	**6.0 s**	**86**
Plain model, no helpers	49/50	6.5 s	68

- Same accuracy as the paper's setup.
- Ours writes 3.5× faster than the paper's way, and faster than either alternative.

"Words" throughout means the word pieces the model writes, roughly three quarters of an English word each.

Many questions at once

Same 50 math questions, sent in groups. Total words written per second, our engine:



13× more output per second when all 50 run together, with 49/50 still correct. The paper's setup cannot do this at all.

Memory

how it was run	GPU memory used
Authors' two-program version	91 GB
Ours (one engine)	**63 GB**

- The difference is mostly the second copy of the model (about 28 GB) that we no longer load.
- That memory can go to serving more questions at once instead.

Summary

- All of LatentMAS now runs inside one fast engine, with one copy of the model.
- Same accuracy on grade-school math as the paper's setup.
- 3.5× faster writing one question at a time.
- 13× more output per second with 50 questions at once.
- About 28 GB less memory than the authors' two-program version.